may one day convince us that, indeed, there is a limit to scientific explanation. But it is also possible, to the contrary, that they will usher in a new era—an era in which we can declare that a fundamental explanation of the universe has finally been found.

Reaching for the Stars

Although we are technologically bound to the earth and its immediate neighbors in the solar system, through the power of thought and experiment we have probed the far reaches of both inner and outer space. During the last hundred years in particular, the collective effort of numerous physicists has revealed some of nature's best-kept secrets. And once revealed, these explanatory gems have opened vistas on a world we thought we knew, but whose splendor we had not even come close to imagining. One measure of the depth of a physical theory is the extent to which it poses serious challenges to aspects of our worldview that had previously seemed immutable. By this measure, quantum mechanics and the theories of relativity are deep beyond anyone's wildest expectations: Wave functions, probabilities, quantum tunneling, the ceaseless roiling energy fluctuations of the vacuum, the smearing together of space and time, the relative nature of simultaneity, the warping of the spacetime fabric, black holes, the big bang. Who could have guessed that the intuitive, mechanical, clockwork Newtonian perspective would turn out to be so thoroughly parochial—that there was a whole new mind-boggling world lying just beneath the surface of things as they are ordinarily experienced?

But even these paradigm-shaking discoveries are only part of a larger, all-encompassing story. With solid faith that laws of the large and the small should fit together into a coherent whole, physicists are relentlessly hunting down the elusive unified theory. The search is not over, but through superstring theory and its evolution into M-theory, a cogent framework for merging quantum mechanics, general relativity, and the strong, weak, and electromagnetic forces has finally emerged. And the challenges these developments pose to our previous way of seeing the world are monumental: loops of strings and oscillating globules, uniting all of creation into vibrational patterns that are meticulously executed in a universe with numerous hidden dimensions capable of undergoing extreme contortions in which their spatial fabric tears apart and then repairs itself. Who could have guessed that the